Qingcheng Zeng

Ph.D. Candidate in Computational Linguistics · Northwestern University

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Summary

Ph.D. candidate working on large language models, with industry research experience at Snowflake, Tencent AI Lab, and Alibaba International. My research spans:

- **LLM Agents, Information Retrieval, and Reinforcement Learning** — search agents, process reward modeling, and retrieval robustness.
- **Trustworthy LLMs** — uncertainty quantification, calibration, and verbalized confidence across LLMs, reasoning models, vision–language models, and tool-use agents.
- **LLMs for Computational Social Science and Social Good** — including public-health and clinical NLP.
- **Understanding LLMs through Cognitive Science.**
- **Multilingual NLP and Machine Translation.**

Industry Experience

Applied Scientist Intern, Alibaba International — Accio Agent Team — June 2026 – Sep 2026
Working with Kunyu Shi and Junbo Li on long-horizon agentic reinforcement learning: trained agents with GRPO and SAO (Single-Rollout Asynchronous Optimization) and helped deliver Occamy-1.0.
Research output: Occamy-1.0: Open Pareto-frontier 35B Intelligence for Co-work, arXiv:2609.11977, 2026.

Research Scientist Intern, Snowflake — Cortex Search AI Team, Remote — March 2026 – June 2026
Working with Puxuan Yu and Aman Mehta.
Research output: SQL schema retrieval at arXiv 2026.

Research Scientist Intern, Snowflake — Cortex Search AI Team, Menlo Park, CA — Sep 2025 – Dec 2025
Working with Puxuan Yu and Aman Mehta.
Shipped instruction-following information retrieval for Cortex Search; research output accepted to EMNLP 2026 (Main).

Research Scientist Intern, Tencent AI Lab America, Bellevue, WA — June 2025 – Aug 2025
Working with Linfeng Song, Xiaoyang Wang, Hongming Zhang, and Wenhao Yu on process reward modeling for agents.
Research output: agent optimization at Findings of ACL 2026.

Education

Ph.D. in Computational Linguistics — 2022–2026 (expected)
Department of Linguistics, Northwestern University, Evanston, IL
Advisors: Professor Rob Voigt and Professor Kaize Ding

B.A. in Linguistics — 2019–2021
University of Manchester, Manchester, UK

B.A. in Linguistics — 2017–2021
Zhejiang University, Hangzhou, China

Academic Research Experience

Visiting Graduate Student, WestlakeNLP, Westlake University — Sep 2024 – Nov 2024
Research in multilingual calibration, supervised by Professor Yue Zhang and Dr. Leyang Cui.
Research output: reasoning-model calibration at EMNLP 2025; LLM evaluation at NeurIPS 2025.

Visiting Graduate Student, Johns Hopkins University (CLSP) Summer 2024
Research in AI-curated democratic discourse at the Center for Language and Speech Processing, supervised by Professor Jason Eisner.
Research output: CrossWeave: Bridging Perspectives Across Online Communities with a Dual-Pane Design, *CSCW 2026* (Demo).

Research Assistant in Natural Language Processing, Zhejiang University, Hangzhou, China 2021–2022
Research in multilingual NLP and biomedical NLP, supervised by Professor Jie Yang.
Research output: effective language model adaptation at IJCAI 2023.

Publications and Preprints

** Equal contribution; # Corresponding author; ‡ Co-second authors.*

LLM Agents, Information Retrieval, and Reinforcement Learning

[1] Dingyuan Dai*, Heli Qi*, Lei Liu*, …, **Qingcheng Zeng**, …, Tianyu Liu*#. OSWorld-Science: A Benchmark of Computer Use Agents for Learning and Using Scientific Software. *arXiv:2609.39903*, 2026.

[2] Weihao Xuan*, **Qingcheng Zeng***, Jiarui Liu*, Rui Yang*, …, Mona Diab. Learning from Use: Test-Time Learning in Large Language Models and Agents. *SSRN Preprint 7319678*, 2026.

[3] Tianyu Liu*, Dingyuan Dai*, Weihao Xuan*, Jialin Chen*, **Qingcheng Zeng***, Rui Yang*, Ada Fang, Zhen Yang, Peng Cui, Yinpeng Dong, Yuanqi Du, Marinka Zitnik, James Zou, Jie Tang. Towards an Agentic Era of AI. *SSRN Preprint 7245280*, 2026.

[4] Wenhui Chen, Shiwen Cheng, Hao Dong, …, **Qingcheng Zeng**, …, Zijian Zou. Occamy-1.0: Open Pareto-frontier 35B Intelligence for Co-work. *arXiv:2609.11977*, 2026.

[5] Zeqi Zhou*, **Qingcheng Zeng***, Rui Yang, Jiarui Liu, Heli Qi, Puxuan Yu, Weihao Xuan#, Naoto Yokoya. When Queries Crowd: Diagnosing and Improving Spread-Out Regularization for Text Embeddings. *EMNLP 2026 (Main)*.

[6] Weihao Xuan*, **Qingcheng Zeng***, Heli Qi, Yunze Xiao, Junjue Wang, Naoto Yokoya#. The Confidence Dichotomy: Analyzing and Mitigating Miscalibration in Tool-Use Agents. *ACL 2026* (Oral).

[7] **Qingcheng Zeng***, Yuheng Lu*, Zeqi Zhou, Heli Qi, Puxuan Yu, Fuheng Zhao, Hitomi Yanaka, Weihao Xuan#, Naoto Yokoya. Code-Switching Information Retrieval: Benchmarks, Analysis, and the Limits of Current Retrievers. *Findings of ACL 2026*.

[8] Mukai Li, **Qingcheng Zeng**, Tianqing Fang, Zhenwen Liang, Linfeng Song, Qi Liu, Haitao Mi, Dong Yu. Verified Critical Step Optimization for LLM Agents. *Findings of ACL 2026*.

[9] Minghao Guo, **Qingcheng Zeng**, Xujiang Zhao, Yanchi Liu, Wenchao Yu, Mengnan Du, Haifeng Chen, Wei Cheng#. DeepSieve: Information Sieving via LLM-as-a-Knowledge-Router. *Findings of EACL 2026*.

[10] Haoyu Wu*, **Qingcheng Zeng***, Kaize Ding#. Fact or Facsimile? Evaluating the Factual Robustness of Modern Retrievers. *CIKM 2025*.

[11] Yuheng Lu*, **Qingcheng Zeng***, Heli Qi, Puxuan Yu, Fuheng Zhao, Rui Yang, Hitomi Yanaka, Naoto Yokoya, Weihao Xuan#. Beyond Monolingual Deep Research: Evaluating Agents and Retrievers with Cross-Lingual BrowseComp-Plus. *arXiv:2606.15345*, 2026.

[12] **Qingcheng Zeng**, Puxuan Yu, Aman Mehta, Fuheng Zhao, Rajhans Samdani. Dual-View Training for Instruction-Following Information Retrieval. *EMNLP 2026 (Main)*.

[13] **Qingcheng Zeng**, Puxuan Yu, Aman Mehta, Fuheng Zhao, Rajhans Samdani. Finding the Right Tables and Columns: A Benchmark and Corpus-Adaptive Embeddings for SQL Schema Retrieval. *arXiv:2607.13311*, 2026.

[14] Fang Wu*#, Aaron Tu*, Weihao Xuan*, Heli Qi*, Xu Huang, **Qingcheng Zeng**, …, Yejin Choi#. Position: The Hidden Costs and Measurement Gaps of Reinforcement Learning with Verifiable Rewards. *arXiv:2509.21882*, 2025.

[15] Shunyu Liu, Minghao Liu, Huichi Zhou, …, **Qingcheng Zeng**, …, Dacheng Tao#. VeriWeb: Verifiable Long-Chain Web Benchmark for Agentic Information-Seeking. *arXiv:2508.04026*, 2025.

Trustworthy LLMs and Uncertainty Quantification

[16] **Qingcheng Zeng***#, Weihao Xuan*, Leyang Cui, Rob Voigt. Thinking Out Loud: Do Reasoning Models Know When They're Right? *EMNLP 2025*.

[17] Weihao Xuan*, **Qingcheng Zeng***, Heli Qi, Junjue Wang, Naoto Yokoya#. Seeing is Believing, but How Much? A Comprehensive Analysis of Verbalized Calibration in Vision-Language Models. *EMNLP 2025* (Oral).

[18] **Qingcheng Zeng***, Mingyu Jin*, Qinkai Yu, Zhenting Wang, Wenyue Hua, Guangyan Sun, Yanda Meng, Shiqing Ma, Qifan Wang, Felix Juefei-Xu, Fan Yang, Kaize Ding#, Ruixiang Tang, Yongfeng Zhang. Uncertainty Quantification for Multiple-Choice Questions is Just One-Token Deep. *CIKM 2025*.

[19] Sijia Chen*, Xiaomin Li*#, Mengxue Zhang, Eric Hanchen Jiang, **Qingcheng Zeng**, Chen-Hsiang Yu. CARES: Comprehensive Evaluation of Safety and Adversarial Robustness in Medical LLMs. *NeurIPS 2025, Datasets & Benchmarks Track*.

[20] Shulin Huang, Linyi Yang#, Yan Song, Shuang Chen, Leyang Cui, Ziyu Wan, **Qingcheng Zeng**, Ying Wen, Kun Shao, Weinan Zhang, Jun Wang, Yue Zhang#. ThinkBench: Dynamic Out-of-Distribution Evaluation for Robust LLM Reasoning. *NeurIPS 2025, Datasets & Benchmarks Track*.

[21] Jiayu Liu*, Rui Wang*, Qing Zong, Yumeng Wang, Cheng Qian, **Qingcheng Zeng**, Tianshi Zheng, Haochen Shi, Dadi Guo, Baixuan Xu, Chunyang Li, Yangqiu Song. NOVA: Noise-Aware Verbal Confidence Calibration for Robust Large Language Models in RAG Systems. *Findings of EMNLP 2026*.

LLMs for Computational Social Science and Social Good

[22] Fei Fang, Reva Hirave, William Jurayj, …, **Qingcheng Zeng**, …, Jason M Eisner. CrossWeave: Bridging Perspectives Across Online Communities with a Dual-Pane Design. *CSCW 2026 Companion* (Demo), pp. 540–545.

[23] Lei Liu*, Zhaokang Liang*, **Qingcheng Zeng**, Chenda Duan, Lu Mi, Zhen Tan, Tianyu Liu#. MERID: Multimodal Exploration via Recursive Self-Improvement Agents for Major Depression Analysis. *arXiv:2609.36235*, 2026.

[24] **Qingcheng Zeng***#, Guanhong Liu*, Zhaoqian Xue, Diego Ford, Rob Voigt, Loni Hagen, Lingyao Li#. Sympathy over Polarization: A Computational Discourse Analysis of Social Media Posts about the July 2024 Trump Assassination Attempt. *Findings of AACL 2025*.

[25] Grace LeFevre, **Qingcheng Zeng**, Adam Leif, Jason Jewell, Denis Peskoff, Rob Voigt. Good Intentions Beyond ACL: Who Does NLP for Social Good, and Where? *EMNLP 2025*.

[26] **Qingcheng Zeng**, Mingyu Jin, Rob Voigt. Adaptive Axes: A Pipeline for In-domain Social Stereotype Analysis. *EMNLP 2024*.

[27] Zhaoqian Xue*, Guanhong Liu*, Chong Zhang, Kai Wei, **Qingcheng Zeng**, Songhua Hu, Wenyue Hua, Lizhou Fan, Yongfeng Zhang, Lingyao Li#. Toward Equitable Access: Leveraging Crowdsourced Reviews to Investigate Public Perceptions of Health Resource Accessibility. *International Conference on Social Computing (ICSC 2025)* (Oral).

[28] Joseph C.Y. Lau, Emily Landau, **Qingcheng Zeng**, Ruichun Zhang, Stephanie Crawford, Rob Voigt#, Molly Losh#. Pre-trained artificial intelligence language model represents pragmatic language variability central to autism and genetically related phenotypes. *Autism*, 29(5):1346–1358, 2025.

[29] Rui Yang*, **Qingcheng Zeng***, Keen You*, Yujie Qiao*, Lucas Huang, Chia-Chun Hsieh, Benjamin Rosand, Jeremy Goldwasser, Amisha Dave, Tiarnan Keenan, Yu He Ke, Chuan Hong, Nan Liu, Emily Chew, Dragomir Radev, Zhiyong Lu, Hua Xu, Qingyu Chen, Irene Li#. Ascle—A Python Natural Language Processing Toolkit for Medical Text Generation: Development and Evaluation Study. *Journal of Medical Internet Research (JMIR)*, 26:e60601, 2024.

[30] Jiageng Wu, Xiaocong Liu, Minghui Li, Wanxin Li, Zichang Su, Shixu Lin, Lucas Garay, Zhiyun Zhang, Yujie Zhang, **Qingcheng Zeng**, Jie Shen, Changzheng Yuan, Jie Yang#. Clinical Text Datasets for Medical Artificial Intelligence and Large Language Models — A Systematic Review. *NEJM AI*, 1(6), 2024.

[31] Mourad Heddaya, **Qingcheng Zeng**, Alexander Zentefis, Rob Voigt, Chenhao Tan. Causal Micro-Narratives. *6th Workshop on Narrative Understanding (WNU), EMNLP 2024*.

[32] Rui Yang, Haoran Liu, Edison Marrese-Taylor, **Qingcheng Zeng**, Yu He Ke, Wanxin Li, Lechao Cheng, Qingyu Chen, James Caverlee, Yutaka Matsuo, Irene Li. KG-Rank: Enhancing Large Language Models for Medical QA with Knowledge Graphs and Ranking Techniques. *BioNLP Workshop, ACL 2024*.

[33] Rui Yang, Huitao Li, Weihao Xuan#, …, **Qingcheng Zeng**, …, Nan Liu#. Toward Global Large Language Models in Medicine. *arXiv:2601.02186*, 2026.

[34] Mourad Heddaya, Chenhao Tan, Rob Voigt, **Qingcheng Zeng**, Alexander Zentefis. A Century of Inflation Narratives. *SSRN Working Paper 5259107*, 2025.

[35] Jiacheng Liu*#, Francesco Salvi*, Philipp Schoenegger*, …, **Qingcheng Zeng**, …, Ezra Karger. When Large Language Models are More Persuasive Than Incentivized Humans, and Why. *arXiv:2505.09662*, 2025.

Understanding LLMs: Cognitive Science and Interpretability

[36] Suet-Ying Lam*#, **Qingcheng Zeng***, Rob Voigt. Encoder-only and decoder-only models understand abstract argument structure differently. *EMNLP 2026 (Main)*.

[37] Ruoxi Ning, Yongpeng Zhu, **Qingcheng Zeng**, Tatsuki Kuribayashi, Freda Shi. On the Effect of Hyperparameters in Language Modeling for Computational Linguistics. *ACL 2026*.

[38] Kefan Yu*, **Qingcheng Zeng***#, Weihao Xuan, Wanxin Li, Jingyi Wu, Rob Voigt. The Pragmatic Mind of Machines: Tracing the Emergence of Pragmatic Competence in Large Language Models. *EACL 2026* (Oral).

[39] Suet-Ying Lam*, **Qingcheng Zeng***#, Jingyi Wu, Rob Voigt. Leveraging Human Production-Interpretation Asymmetries to Test LLM Cognitive Plausibility. *ACL 2025* (Short Paper).

[40] Mingyu Jin*, Qinkai Yu*, Jingyuan Huang*, **Qingcheng Zeng***, Zhenting Wang, Wenyue Hua, Haiyan Zhao, Kai Mei, Yanda Meng, Kaize Ding, Fan Yang, Mengnan Du, Yongfeng Zhang#. Exploring Concept Depth: How Large Language Models Acquire Knowledge and Concept at Different Layers? *COLING 2025*.

[41] Suet-Ying Lam*, **Qingcheng Zeng***#, Kexun Zhang*, Chenyu You, Rob Voigt. Large Language Models Are Partially Primed in Pronoun Interpretation. *Findings of ACL 2023*.

[42] **Qingcheng Zeng**, An-Ran Li. A Survey in Automatic Irony Processing: Linguistic, Cognitive, and Multi-X Perspectives. *COLING 2022*.

[43] Mingyu Jin, Yutong Yin, Jingcheng Niu, **Qingcheng Zeng**, Wujiang Xu, Xinyuan Song, Wei Cheng, Mengnan Du, Zhaoran Wang, Tianlong Chen, Dimitris N. Metaxas. Harder the Task, Sparser the Representation: Sparsity as a Learning Signature of Capability in LLMs. *NeurIPS 2026*.

[44] Ziqi Gao, Yaotian Zhu, **Qingcheng Zeng**, Xu Zhao, Ziqing Wang, Feng Ruan, Kaize Ding. RAPTOR: Ridge-Adaptive Logistic Probes. *NeurIPS 2026*.

Multilingual NLP and Machine Translation

[45] Xinhe Shi, **Qingcheng Zeng‡**, Weihao Xuan‡, Linchao Zhu#. How to Improve LLMs' Performance on Specific Languages: A Perspective on LLM-Derived Language Similarity. *ACL 2026* (Oral).

[46] Weihao Xuan, Rui Yang, Heli Qi, **Qingcheng Zeng**, …, Irene Li. MMLU-ProX: A Multilingual Benchmark for Advanced Large Language Model Evaluation. *EMNLP 2025*.

[47] Daoyang Li, Haiyan Zhao, **Qingcheng Zeng**, Mengnan Du. Exploring Multilingual Probing in Large Language Models: A Cross-Language Analysis. *XLLM Workshop, ACL 2025*.

[48] **Qingcheng Zeng***, Lucas Garay*, Peilin Zhou*, Dading Chong, Yining Hua, Jiageng Wu, Yikang Pan, Han Zhou, Rob Voigt, Jie Yang#. GreenPLM: Cross-Lingual Transfer of Monolingual Pre-Trained Language Models at Almost No Cost. *IJCAI 2023*.

Other Publications

[49] Fan Gao, Hang Jiang, Rui Yang, **Qingcheng Zeng**, Jinghui Lu, Moritz Blum, Dairui Liu, Tianwei She, Yuang Jiang, Irene Li. Evaluating Large Language Models on Wikipedia-Style Survey Generation. *Findings of ACL 2024*.

[50] Dading Chong, Helin Wang, Peilin Zhou, **Qingcheng Zeng**. Masked Spectrogram Prediction for Self-Supervised Audio Pre-Training. *ICASSP 2023*.

[51] **Qingcheng Zeng**, Dading Chong, Peilin Zhou, Jie Yang#. Low-resource Accent Classification in Geographically-proximate Settings: A Forensic and Sociophonetics Perspective. *Interspeech 2022*.

[52] Peilin Zhou, Dading Chong, Helin Wang, **Qingcheng Zeng**. Calibrate and Refine! A Novel and Agile Framework for ASR Error Robust Intent Detection. *Interspeech 2022*.

[53] Sheng Jin*, Minghao Liu*#, Yunze Xiao, …, **Qingcheng Zeng**, …, Ge Zhang#. Knowledge Index of Noah's Ark. *arXiv:2606.05104*, 2026.

Honors & Awards

- Gold Reviewer, ICML 2026
- Excellent Reviewer, KDD 2025 Applied Data Science Track
- TGS Conference Travel Grant (EMNLP 2024, $800), Northwestern University
- Departmental Travel Grant (EMNLP 2024, $400), Northwestern University
- TGS Conference Travel Grant (IJCAI 2023, $850), Northwestern University
- Graduate Fellowship, Northwestern University
- Q-Step Center Best Dissertation Award (£100), University of Manchester
- The Dean Award for Achievement, University of Manchester

Invited Talks

On Verbalized Confidence and Factuality, LARK Lab, HKUST (Guangzhou) — Oct 2025

Teaching Experience

Teaching Assistant, LING 220: Language & Society, Northwestern University — Spring 2025
Teaching Assistant, LING 331: Text Processing for Linguists, Northwestern University — Winter 2025
Teaching Assistant, COGSCI 110: Introduction to Cognitive Science, Northwestern University — Fall 2023

Academic Service

Conference reviewer: ACL Rolling Review, NeurIPS, ICML, COLM, KDD, WWW, IJCAI, IC2S2, ICASSP

Journal reviewer: Artificial Intelligence; IEEE Journal of Selected Topics in Signal Processing; IEEE Transactions on Neural Networks and Learning Systems (TNNLS); Computer Speech and Language; Knowledge-Based Systems; Expert Systems With Applications; IEEE Access; Information Processing and Management; Neurocomputing; Neural Networks; Pattern Recognition; Applied Soft Computing; JMIR AI; JMIR Medical Informatics; International Journal of Medical Informatics; Telematics and Informatics Reports; Acta Psychologica

Departmental service: Colloquium Committee Member, Northwestern Linguistics — 2023–2024

References

Rob Voigt (Ph.D. Advisor)
Assistant Professor of Linguistics
University of California, Davis
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Kaize Ding (Ph.D. Co-advisor)
Assistant Professor of Statistics and Data Science
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Lingyao Li (Ph.D. Committee)
Assistant Professor of Information
University of South Florida
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Matt Goldrick (Ph.D. Committee)
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matt-goldrick@northwestern.edu